ARJUN GUPTA

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EDUCATION

Kanpur Institute of Technology: Bachelor of Computer Applications *[GPA: 7.22/10]* **Expected 2028**
Relevant Coursework: Linear Algebra, Calculus, Data Structures & Algorithms, Machine Learning, Software Engineering

RESEARCH EXPERIENCE

Research Intern: NLP / RAG **April 2026 – July 2026**
IIT Bhubaneswar

- Implemented and benchmarked advanced RAG chunking strategies based on the KET-RAG paper.
- Reproduced and benchmarked DynaBERT-based transformer compression pipelines under fixed hardware and latency budgets.
- Width and depth-adaptive compression at 0.75× on RoBERTa-base yielded a 21.3% smaller model with a 1.78× inference speedup, while retaining 94.6% of baseline accuracy on QQP.

ML Intern: Machine Learning in FinTech **November 2026 – December 2026**
IIT Roorkee

- Developing data extraction pipelines from Bloomberg and GDELT at the Financial Analytics Lab (Dept. of Management Studies, under Prof. Gaurav Dixit) to feed downstream analysis and NLP tasks.
- Built a Python/blpapi Bloomberg CLI querying 240+ fields per security; pipelined concurrent requests to cut retrieval time 5.9× (7.6s → 1.3s), reaching ~6,100 data points/s.

PROJECTS

NeuroVision: EEG to Image Retrieval **October 2026**
Technologies: *PyTorch, Scipy, EEG, ResNet-50*

- Built and evaluated a CNN EEG encoder aligned to frozen ResNet-50 features with contrastive InfoNCE loss on the THINGS-EEG2 dataset. It reached 18.6% top-1 and 43.8% top-5 zero-shot retrieval among 200 unseen-concept images.
- It outperformed a ridge-regression baseline (14.2% top-1) by 4.5 points. A time-window analysis showed the signal is concentrated in the first 400 ms: 400–790 ms gives 1.5% top-1, against 9.8% for 0–200 ms and 11.9% for 200–400 ms.

Industrial Anomaly Detection: PatchCore **March 2026**
Technologies: *Python, PyTorch, WideResNet50*

- Reproduced the PatchCore anomaly detection paper on the MVTec AD dataset, achieving 0.99 AUC using WideResNet50 feature extraction.
- Implemented coreset subsampling and KNN-based anomaly scoring from scratch, reducing the memory bank size by 90%+ with negligible loss in detection accuracy.

Orbital-X: Exoplanet Radial-Velocity Orbit Models **August 2026**
Technologies: *PyTorch, Astropy, Scipy, PINN*

- Built a PyTorch pipeline benchmarking least-squares, a PINN and a Neural ODE for exoplanet RV orbit fitting: ~6,000 batched, checkpointed fits plus 10 automated correctness checks.
- On 40 held-out systems, the classical fit reconstructed the true RV curve 0.25 m/s more accurately than both neural methods (p=0.0001) and beat the PINN on held-out RMSE (2.19 vs 2.84 m/s, p=0.003). Parameter recovery was statistically tied (95% vs 92.5%/90%), and classical was 5–70× cheaper (amortised runtime).

Jal Rakshak: Smart Water Tank Overflow Detection **June 2026**
Technologies: *TensorFlow, YAMNet, Flutter, Android SDK*

- Built a real-time water tank overflow detection system that classifies tank-fill sounds captured via a wireless mic, trained on a self-recorded custom dataset of 40 minutes of tank-fill audio collected at home.
- Deployed the model on Android and Windows with an auto-train feature that retrains on user-labeled data, allowing accuracy to improve continuously post-deployment.

OPEN SOURCE CONTRIBUTIONS

- **Merged Pull Request**: ml-recipes-py: Contributed a standardized ML pipeline featuring structured EDA, automated feature preprocessing, and Logistic Regression modeling to establish baseline code patterns and documentation standards.

TECHNICAL SKILLS

Languages: Python, C
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Pandas
Scientific Computing: NumPy, SciPy, Astropy, TorchDiffEq

Concepts: Transfer Learning, CNNs, Anomaly Detection, Transformers, Neural ODE, Physics Informed ML, Natural Language Processing (NLP), Data Structures and Algorithms
Tools: Jupyter, Git, GitHub, VS Code

INTERNATIONAL

JST Sakura Science Exchange Program | Iwate Prefectural University, Japan

- Selected by host university for a JST-funded research exchange in advanced computing.
- Fully funded 1 week visit to Prof. Hirokazu Madokoro's Computer Vision Lab in January 2027.

Vision Transformer Benchmarking (OutcropHybNet Replication)
Collaborative Study with Prof. Hirokazu Madokoro
Technologies: *PyTorch, timm, Hugging Face, SegFormer, DeepLabv3+*

- Validated the core empirical finding that vision transformers outperform traditional CNNs: SegFormer MiT-B1 achieved 36.67% mIoU compared to 33.76% for DeepLabV3+ (ResNet-101) and 28.26% for DeepLabV3 (MobileNetV3), outperforming ResNet-101 on 108/169 classes (64%) and MobileNetV3 on 137/169 classes (81%).
- SegFormer required ~3.5× fewer parameters (13.7M vs. 48.1M), 3.6× fewer GFLOPs (31.9 vs. 115.5), 1.4× lower inference latency (12.2 ms vs. 17.6 ms), and 45% less training time compared to DeepLabV3+ with ResNet-101.

COURSES

- Machine Learning Specialization: Stanford University
- Deep Learning for Computer Vision and NLP: IIT Guwahati
- Introduction to Computational Neuroscience: MIT OpenCourseWare
- Introduction to Astrophysics: EPFL